# Andrews Acheampong

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Andrews Acheampong | AcheampongAndy

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## Research Interests

Bayesian Statistical Methods, Scalable MCMC, Stochastic Modelling, Mathematical Biology, Computational Statistics, and Statistical Genomics.

## Education

- **Kwame Nkrumah University of Science and Technology [🌐]** *January 2026 - Present*
  *MPhil. Biodata Analytics and Computational Genomics* Kumasi, Ghana
  - First Semester: WA 72.33/100; 18 credits completed.
  - Relevant coursework: Modern Statistical Methods, Mathematical Modeling in Biology, Genomes and Gene Expression and Advanced Scientific and Technical Computing.
- **Kwame Nkrumah University of Science and Technology [🌐]** *January 2021 - August 2024*
  *Bachelor of Science in Statistics* Kumasi, Ghana
  - CWA: 74.49/100; 154 credits completed.
  - Class Rank: Top 5% of graduating class (240 students).

## Research Experience

- **Network-Informed Bayesian Shrinkage Model for Ordinal Biological Outcomes** *2026 - Present*
  *MPhil Research Project, Supervisor: Dr. Isabel Mensah, Department of Mathematics, KNUST* Kumasi, Ghana
  - Developing a Bayesian model for ordered biological outcomes with high-dimensional genomic predictors.
  - Combining ordinal regression, Bayesian shrinkage, and gene-network information to account for relationships among genes, improve prediction, and identify important predictors.
  - Planning simulation studies and application to an existing biological dataset to evaluate predictive performance, variable selection, and uncertainty quantification.
- **Multidrug-Resistant Bacteria in Catfish Farming Systems** *2026*
  *Research Study, Kumasi Veterinary Laboratory* Kumasi, Ghana
  - Analyzed fish and water data collected from 16 aquaculture farms using R, summarizing bacterial occurrence, antimicrobial susceptibility patterns, and multidrug resistance.
  - Used frequencies, percentages, confidence intervals, tables, and figures to compare resistant bacteria across sample types and support biological interpretation.
  - Manuscript from the study is currently being prepared for publication.
- **Genetic and Statistical Foundations of Coalescent Theory and Its Applications in Population Genetics [🌐]** *July 2026*
  *Graduate Term Paper*
  - Examined coalescent theory using the Wright–Fisher model, stochastic processes, and backward-time population genetic inference.
  - Studied Watterson's estimator, nucleotide diversity, the site-frequency spectrum, and Tajima's $D$ for inference on effective population size, demographic history, migration, selection, and ancestry.
- **Predictive Analytics for Kidney Disease Progression [🌐]** *2024*
  *Undergraduate Thesis*
  - Developed a logistic regression model to investigate clinical factors associated with chronic kidney disease and evaluated performance using 10-fold cross-validation.
  - Achieved 97% classification accuracy and a Kappa value of 0.94.

## Laboratory Experience

- **Kumasi Veterinary Laboratory, Molecular Department** *May 4-22, 2026*
  *Wet Laboratory Intern, Molecular Diagnostics* Kumasi, Ghana
  - Completed supervised training in DNA/RNA extraction, PCR master-mix preparation, PCR-based viral detection, amplification-output interpretation, contamination-aware sample handling, and laboratory documentation.

## Academic and Teaching Experience

- **Data Science for Chinese American Biopharmaceutical Society (DS4CABS) [🌐]** *October 2026 – Present*
  *Mentor* Remote
  - Appointed as a mentor in the DS4CABS program, an initiative connecting learners and professionals in data science and biopharmaceutical research.
- **Department of Statistics and Actuarial Science, KNUST [🌐]** *October 2024 - October 2025*
  *Research and Teaching Assistant* Kumasi, Ghana
  - Conducted tutorials in Probability Distributions and prepared learning activities in Probability and Measure Theory and Probability and Statistical Inference.
  - Assisted faculty research in data science and machine learning and supported final-year undergraduate students.
- **Independent Statistics and R Instruction** *September 2024 - July 2025*
  *Instructor* Remote
  - Taught data analysis with R, including data manipulation, visualization, statistical analysis, and applications to medical datasets.

## Bioinformatics and Computational Projects

- **Rhinovirus Classification Web App** *January 2025*
  *Python, Flask, Biopython, SQLite, Matplotlib* [🐙]
  - Developed a web application for FASTA sequence analysis, Rhinovirus classification, and phylogenetic visualization.
- **Analysing and Interpreting Genomics Datasets**
  *Linux, Bash, R, NGS workflows, Nextflow concepts* [🐙]
  - Built a practical genomics repository covering sequencing file formats, quality control, mapping, variant-calling workflows, and reproducible command-line analysis.
- **Bioinformatics for Biologists**
  *Bash, Linux, biological data workflows* [🐙]
  - Completed command-line bioinformatics exercises focused on reproducible biological-data handling and analysis workflows.

## Selected Presentations and Workshops

- **Mathematical Biology Training Workshop** *October 19–23, 2026*
  *AIMS Ghana Research Centre, University of Ghana*
  - Participated in intensive training in mathematical biology, covering biological modelling, epidemiology, immunology, and Python-based computational methods.
  - Engaged in research seminars, practical exercises, and collaborative group project activities.
- **Data-Driven Decision – Building the Future We Want with Statistics** *February 12, 2025*
  *Ghana Association of Statistics Students, KNUST*
  - Delivered a keynote address on the role of statistics in data-driven decision-making, research, policy, and business intelligence.
- **Introduction to Git and GitHub for Research Collaboration** *November 19–21, 2024*
  *Department of Statistics and Actuarial Science, KNUST*
  - Conducted hands-on training for graduate and research assistants on repositories, commits, branches, pull requests, and collaborative version control.

## Computing and Research Skills

- **Statistical Computing:** R, Python, SQL; regression, cross-validation, Bayesian modeling, data visualization
- **Bioinformatics:** Linux, Bash, FASTA/FASTQ workflows, sequence quality control, mapping and variant-calling concepts, Biopython
- **Research Tools:** Git, GitHub, Conda, reproducible workflows, scientific writing and presentation

## Selected Certifications

- **Mathematical Biology Training Workshop – AIMS Ghana** *October 2026*
- **Basic Molecular Biology Module 3** *August 2025*
- **An Introduction to Linux, Bash Scripting, and R** *March 2025*
- **Basic Molecular Biology Module 1** *September 2024*
- **Data Science with R** *April 2024*

## References

Available upon request.